Journal of Urban Economics xxx (xxxx) xxx

Contents lists available at ScienceDirect

Journal of Urban Economics

journal homepage: www.elsevier.com/locate/jue

Measuring urban economic density☆

J. Vernon Henderson*, Dzhamilya Nigmatulina, Sebastian Kriticos

Department of Geography and the Environment, Houghton Street, Greater London, London, WC2A2AE, United Kingdom

ARTICLE INFO

JEL classification:
J24
J31
O15
O55
R1
R3

Keywords:
Cities
Economic density
Agglomeration economies
Urbanization
Pull forces
Sub-Saharan africa
Rural-urban migration
Landscan
Economic development

ABSTRACT

At the heart of urban economics are agglomeration economies, which drive the existence and extent of cities. This paper estimates urban agglomeration effects, exploring simple and very nuanced measures of economic density to explain household income and wage differences across cities in six Sub-Saharan African countries. A key aspect of the work is that we define cities consistently across space based on fine scale density measures, in order to gauge the economic extent of the city. The evidence suggests that more nuanced measures of density, which attempt to capture within-city differences in the extent of clustering, do no better than a simple density measure in explaining income differences across cities. However, total population is a poor measure. We find large wage gains to being in denser cities in Sub-Saharan Africa, generally larger than such estimates for other parts of the world. We also find extraordinary household income gains to density that are far greater than wage ones. Such gains help explain the pull forces driving rapid urbanization in the region.

1. Introduction

At the heart of urban economics are the sources and nature of agglomeration economies, which drive the existence and the extent of cities. The extent of agglomeration economies help determine the role of urban pull factors in the urbanization process within and across regions of the world. Sub-Saharan Africa has had more rapid urban population growth than any other region over the last half century. Its urban population share grew from 14% in 1950 to 41% in 2015 and is predicted to be 60% in 2050 according to the UN (2015). This rapid urbanization is driven by forces pushing people out of rural areas and pulling them into cities as reviewed for Sub-Saharan Africa in Henderson and Kriticos (2018). Push factors include changes in agricultural technology which release workers from land (e.g. Matsuyama (1992); Gollin et al. (2007), and Bustos et al. (2016)) as well as adverse climate and conflict in rural areas (e.g., Henderson et al., 2017; Fay and Opal, 1999; Barrios et al., 2006; Brückner, 2012). With some exceptions such as Gollin et al. (2013), the literature on Sub-Saharan Africa has focused on push factors as driving urbanization. Here, like Gollin et al. (2013), we argue that the income gains associated with moving to cities from rural areas are high in Sub-Saharan Africa, pulling people into cities, albeit in the face of poor productivity in the rural sector. We also show that, within the urban sector, there are huge gains to moving to denser agglomerations.

There is a vast literature that has traditionally focused on developed countries - such as the USA - which estimates the productivity or wage premiums from being in cities compared to rural areas, or from being in bigger versus smaller, or denser versus less dense cities (e.g Ciccone and Hall, 1996; Glaeser and Mare, 2001), with reviews of the literature in Combes and Gobillon (2015) and Rosenthal and Strange (2004). The literature argues that premiums exist because agglomeration gives proximity among firms, workers and people, which allows economic agents to economize on local trade costs, spread information and ideas more easily, diversify the range of products produced, and access larger pools of workers and jobs (Duranton, 2015). Recently, there has been more work on developing countries with, for example, papers on Latin

☆ This work carried out under an Africa Research Program on Spatial Development of Cities at LSE and Oxford which is funded by the Multi Donor Trust Fund on Sustainable Urbanization of the World Bank and by the UK Department for International Development. We thank Patricia Jones for sharing value added, location, employment and industrial code data for Kampala firms from the Uganda Business Inquiry survey of 2002. We thank the editor and two referees for their helpful comments.

* Corresponding author.
E-mail address: J.V.Henderson@lse.ac.uk (J.V. Henderson).

https://doi.org/10.1016/j.jue.2019.103188
Received 30 July 2018; Received in revised form 9 August 2019
Available online xxx
0094-1190/© 2019 Elsevier Inc. All rights reserved.

Please cite this article as: J.V. Henderson, D. Nigmatulina and S. Kriticos, Measuring urban economic density, Journal of Urban Economics, https://doi.org/10.1016/j.jue.2019.103188

America (Quintero and Roberts, 2018), Colombia (Duranton, 2016), China (Combes et al., 2019), and a comparison of Brazil, China, India and the USA by Chauvin et al. (2017).

Wage premiums in developing countries may be larger because their small informal sector firms and poorly educated entrepreneurs and workers are more reliant on their external environment, so that the agglomeration factors and externalities noted above may play a bigger role than in developed countries. Duranton (2015) argues that returns to density will be larger in developing countries. That said, other papers suggest that for Sub-Saharan Africa, cities have low economic density or proximity, resulting in lowered spillovers (Collier and Jones, 2016; Venables, 2018; Lall et al., 2017) and suppressing gains from urbanization. Low economic density arises, for example, if economic activity is not very clustered but scattered throughout the city, potentially because of the high costs of commuting within cities inducing firms to locate nearer to residents as in Fujita and Ogawa (1982), with corresponding empirical work in Heblich et al. (2018). In Appendix D, we examine whether large cities in Sub-Saharan Africa have a lower degree of density and clustering of economic activity relative to the rest of the world. Using LandScan (2012) data, we find that these cities in fact generally have higher density and clustering measures than Latin American cities, and measures that are comparable to Asian cities. That said there is enormous variation in density across Sub-Saharan African cities of similar populations and these density differences drive huge wage and income differences.

Testing the gains from proximity and agglomeration in Sub-Saharan Africa, or elsewhere, involves three critical aspects of measurement, hitherto under-explored in the previous literature. First is how cities are defined, with an eye to establishing consistency in how densities are measured across cities, where density will be a key measure of agglomeration. The main problem concerns the extent of the land area over which a city is defined. The land area and its corresponding population are typically defined by administrative boundaries, rather than economic 'boundaries' related to density. More specifically, the definitions of cities and urbanized areas chosen by national statistics bureaus are typically based on pre-determined administrative units which are defined as either urban or not, based on qualitative aspects of land use and the built environment, such as the degree of centrality of activity, total population and the like. In some countries, population density plays a role in the definition but typically not a central one. Contiguous urban administrative units are then aggregated into a metropolitan or urban area. If land areas are based on administrative units which have populated and unpopulated parts and inconsistently defined across countries or regions within countries, then density measures will fail to capture the *de facto* economic density at which a city's activity operates. Put differently there can be large measurement error leading to attenuation bias, as we will demonstrate.

The second measurement aspect is to determine what specific population and density related measures best capture agglomeration forces, or best capture what we term economic density, the forces driving income and wage premiums. Most studies adopt specific, simple measures of agglomeration such as indicator variables for settlement type (e.g. urban versus rural), a continuous total population measure, or at best, a basic population density measure, where as noted the denominator (effective land area) is typically ill-defined across cities. Studies usually pick one specific measure with little analysis as to why and no quantitative evaluation of what measure(s) would be most relevant.

The third measurement aspect concerns outcomes. Outcomes in the literature are specified either as firm productivity or individual wages, but not household incomes. For rapidly urbanizing regions, in the long term it is more households that ultimately move to cities. If household gains from being in urban areas differ from wage rate gains, that is a critical factor in studying urbanization.

This paper differs in each of these three dimensions. A novel part of our work will be to have consistent city boundaries based on fine scale density measurements, which will reduce measurement error and provide really nice results. We will consistently define metropolitan areas across countries, based on population and population density thresholds. Second, we will focus on an evaluation of different agglomeration aspects starting with traditional options: urban versus rural, a continuous total population measure, and a continuous measure of overall population density. However, these measures do not capture, for a given population and a given overall density, the degree to which economic activity is clustered within a city. Two cities with the same density and population may have very different levels of clustering of economic activity within the city, and they can be captured by other measures reflecting variation in population densities within a city (De La Roca and Puga, 2017; Collier et al., 2018). We derive and utilize several such measures. Third, we will focus not just on wage rates, but, as a novel feature, on household incomes. Agglomeration effects will be much higher for household incomes and we will explore reasons why.

Specifically, the paper will examine how well different economic density measures explain household income and wage differentials across space for a set of six Sub-Saharan Africa countries whose total population is about 430 million. Our sample covers the rural sector, 193 low-density urban settlements with over 5000 people, and 120 high-density cities with over 50,000 people. Using populations at the 1 km grid square level, we aggregate contiguous squares of high density to create cities – which are the consolidation of an urban core and a surrounding lower density fringe – and aggregate contiguous squares of lower density places to create stand-alone, low density settlements, which from now on we will call towns. While admittedly the density and population thresholds we choose below are to some degree arbitrary, they are based on types of thresholds some countries and researchers cite, or use in the case of OECD (2012).

For the measurement of agglomeration economies, there is the issue of determining relevant spatial scales. For example, does it matter whether you are just in a particular city or what part of the city you are in? First we examine how marginal scale effects, or elasticities, vary across the spatial hierarchy: rural areas, towns, and cities. Then for cities, while most papers explore agglomeration benefits at the level of a city or county, a few papers look within cities at the extent of spatial decay, finding a very rapid spatial decay of certain types of scale externalities (Arzaghi and Henderson, 2008; Rosenthal and Strange, 2008). Here we explore both city level and neighborhood effects together: those from overall city economic density and from own local neighborhood economic density. We will ask also if people living in the urban core versus fringe of the city benefit differently from overall density characteristics of the city.

In the paper, Section 1 starts with how we define cities and what are the advantages and disadvantages of the LandScan (2012) data set that we use. Section 2 defines various measures of economic density for urbanized and rural areas, decomposing them into first and second moment components. Section 3 looks at the relationship between different measures of economic density and income differentials across the whole spatial hierarchy. Section 4 looks specifically at cities and examines issues such as the optimal rate of spatial discount for De La Roca and Puga (2017) measures of neighbor effects; the role of spatial variation in density within a city; how important local density measures are within cities, as well as location within the city; and what factors may underlie the huge income gains to families from moving to high density places. Section 5 concludes and discusses extensions.

2. Using landscan data and defining urbanized areas

2.1. Landscan data

To analyze measures of economic density and clustering withing the city, we need information on where people and workers are at a fine spatial resolution. And we need the most accurate data available. For countries like the USA, both finely gridded population and employment data are available from censuses. However, in most developing countries that

is not the case. Population data may only be available at a coarse scale such as regional or local government administrative units and economic censuses in Sub-Saharan Africa are generally non-existent. In terms of population the key then is to allocate people from coarse administrative units to a spatial scale such as a square kilometer.

Our primary data source is LandScan (2012) from Oak Ridge National Laboratory in the USA, which is now being used in some research (e.g. Desmet et al., 2018). Oak Ridge takes population data from sources such as the US Bureau of the Census which assembles census data from other countries on as fine an administrative scale for each country as they can obtain, such as provinces, counties, parishes or even subcounties. Landscan then creates a measure of an ambient population for approximately each 1km grid square on the planet (30"x30" arcseconds). The ambient population is meant to represent where people are on average over the 24 h day. To assess the ambient population, they appear to use nocturnal and diurnal population estimates for at least some areas of the globe, although these are not publicly accessible.

For Landscan, as for WorldPop[1], the Global Human Settlements Layer [2], and similar data sets, a key element in this process involves taking population numbers at some upper level of spatial scale and allocating people to fine grid squares based on where they are likely to live and possibly work. The typical standard in such work has been to allocate people on the basis of ground cover in a grid square from Landsat satellite imagery, or its enhanced versions. This is viewed as an improvement on the Gridded Population of the World, which simply smears population of an administrative unit uniformly across the 1 km grid squares in the unit.

Landscan has two key advantages and two key disadvantages over other data sets. First, Oak Ridge National Lab is more explicit in the fact that they are trying to estimate the ambient population with potentially nocturnal and diurnal populations; while in other algorithms this is implicit through the smearing of the population into general built cover, without workplace or residence distinction. Diurnal is meant to include where the employed population is found during the work day as well as where the non-working population is found; while nocturnal is primarily where people live and sleep. The second advantage of Landscan is that Oak Ridge National Lab has access to very high-resolution satellite data and a wealth of other information which *potentially* allows them to distinguish where built cover is likely to house employment versus residents versus perhaps even shoppers, as well as *potentially* to distinguish roads for commuting and even infer intensity of building on a spot (Rose and Bright, 2014).

A key disadvantage in using Landscan is the complete lack of specificity and transparency as to what Oak Ridge researchers actually do; hence, our use of the word "potentially" in describing what they might do. The second disadvantage, which they acknowledge, is that Landscan data for different time periods are not comparable over time, presumably both because of differential availability of high-resolution data over time and increasingly sophisticated extraction of information from later satellite images.

Hopefully in the future, proposed data sets such as the High Resolution Human Settlement Layer (HRSL)[3] or Modelling and Forecasting African Urban Population Patterns (MAUPP)[4] – which also use very high spatial resolution data – will be able to cover a wider set of time periods and countries consistently with a more explicit methodology. Then one will be able to compare them with Landscan and related data sets and do more comprehensive ground-truthing exercises. Second while ambient population may be a relevant measure to use in characterizing economic density, one would like also to know about clustering and density of employment only. If the assignment of people to workplace buildings gets very sophisticated, we would be able to explore the use of employment density measures, as well as those of ambient population.

Since we will be looking within cities at ambient population density in order to measure the extent of clustering, we want to know if Landscan data are at least somewhat reliable for Sub-Saharan African cities where fine spatial data on population and employment location are generally not available and in many cases do not exist. In Appendix C, we attempt to ground-truth Landscan measures at the grid square level for Kampala and Nairobi, two cities where we have fine spatial resolution data on population and employment which would be unavailable to Oak Ridge National Lab. We further attempt to replicate Landscan's ambient population measure by grid square using an assignment algorithm on our own data. We conclude that Landscan measures do well.

2.2. Defining urbanized areas

As noted earlier, the problem with typical definitions of urbanized areas from the United Nations or population censuses of different countries is that they employ country-specific city and town definitions incorporating administrative boundaries, which means there is no consistency across countries and economic land area is poorly measured. Second, most definitions of "urban" are based to a large extent on qualitative and subjective criteria. Some countries use an application or evaluation process to redefine rural administrative areas as urban, which tends to under-represent newer fast-growing agglomerations due to delays in application, evaluation, and granting of urban status.

To avoid the administrative boundary problem, researchers have taken several approaches. One is to define commuting zones, based on very fine spatial scale data in the USA or EU; such data does not exist in most developing countries. Another is to use night lights such as in Harari (2016) or Henderson and Kriticos (2018). Night lights provide contiguously lit polygons, in which the commuting zone may be included due to light from cars on the roads. Night light data can be used to track the evolution of cities throughout 1992–2013, albeit with some inconsistencies across the sensors. To define the extent of a city one must impose a lights threshold at the boundary with rural areas. Given blooming of lights over unlit areas and more general electrification of the rural population, the major drawback in using night lights data is that one must impose different light thresholds globally. In Sub-Saharan Africa to capture true but smaller urban areas, one can't impose a lights threshold (above 1 on a scale of 0–63) to define cities (see Henderson and Kriticos, 2018), but with no threshold a few unreasonably large urban areas can emerge. India with more widespread electrification requires a high threshold, so that many large separate cities do not merge into one (Harari, 2016). Dingel et al. (2018) attempt to match a night-lights threshold to that of commuting flows in Brazil and apply it to other developing countries such as China. However, this approach works only if the degree of electrification is the same everywhere else as in Brazil, and the commuting flows are the same. It is interesting to note that Harari (2016) uses a threshold of 35 for India, while Dingel et al. (2019) use 30 for Brazil, a more developed country.

A third way to define boundaries of cities is to use the extent of built cover (Angel et al., 2016 and Earth Observation Center at the German Aerospace Center documented in Esch et al., 2018) to name a few), using, typically, Landsat satellite data which is global. Landsat data are based on low-resolution 30m-by-30m satellite data, which has enough bands to distinguish man-made (or impermeable surface) from natural cover (Angel et al., 2016). As an alternative, the German Aerospace Center uses Radar satellite information to infer settlements from the elevation of the ground and structures. The two above-mentioned projects have provided for open use their boundaries of building extent for 1990, 2000 and 2014 in the case of the Marron Institute, and for 2011 in the case of the Earth Observation Center. However neither data provider attempts to calibrate the correct threshold for build cover to match the commuting flows. For example, Marron Institute just picks the threshold of 50% of 30 m × 30 m pixels built within 1km radius.

[1] http://www.worldpop.org.uk/data.
[2] http://ghsl.jrc.ec.europa.eu/datasets.php.
[3] https://www.ciesin.columbia.edu/data/hrsl.
[4] http://spell.ulb.be/project/maupp.

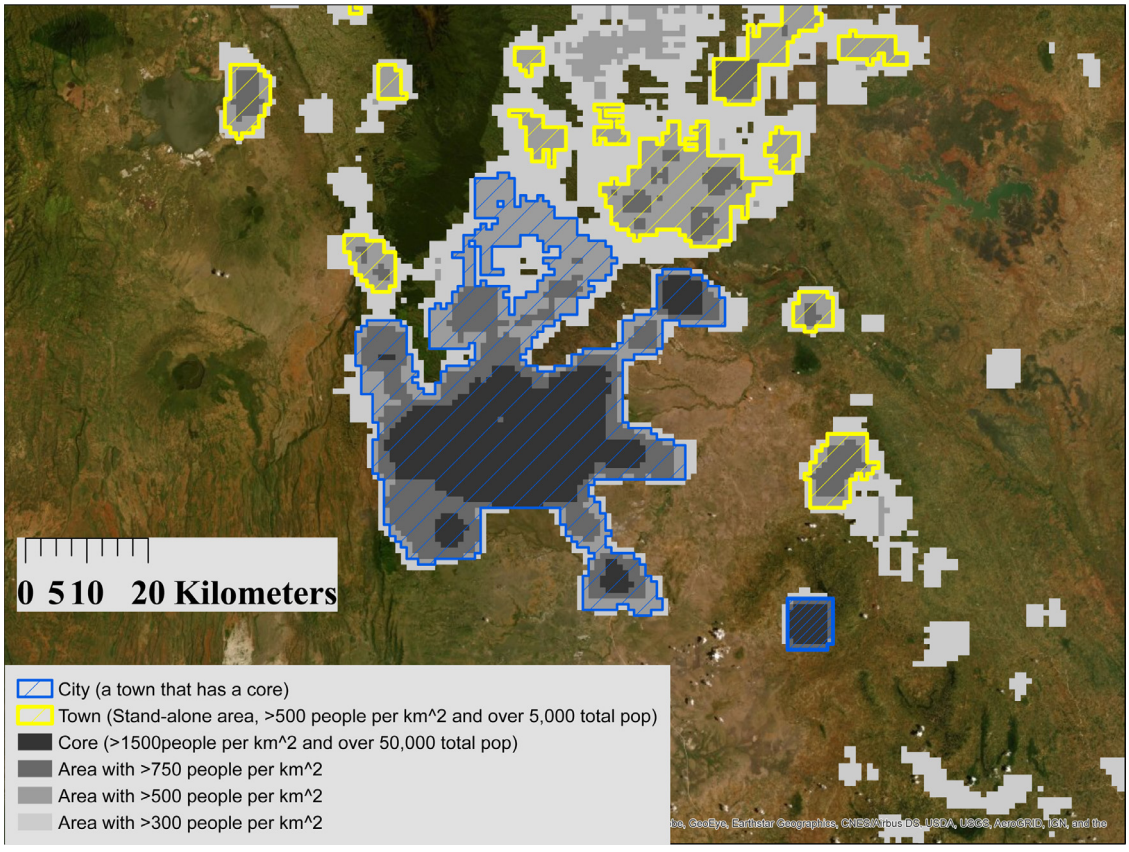


Fig. 1. Defining cities and towns.

Building on this third way, using GHSL data (OECD (2012)), the European Union Commission (https://ghsl.jrc.ec.europa.eu/CFS.php) defines cities based on population density, where, in the GHSL, population from census administrative units is assigned to grid squares based on the degree of built cover. Then the Commission uses thresholds of population (density) of these grid squares to define city boundaries. We follow a similar procedure and employ a consistent density based definition across our African countries, using LandScan (2012) ambient population per grid square. We set density thresholds to define both cities and towns. To do so, we apply a smoothing algorithm as described in Appendix B1 so that each own grid square is assigned the average density of the 7 km x 7 km neighborhood in which it is the center. Smoothing avoids large doughnut holes in cities, due to terrain factors, airfields, parks, big open public spaces and the like. We define a core city as a set of contiguous grid squares all of which have a density greater than or equal to 1500 per sq. km. and the total population of these contiguous squares must sum to 50,000 or more. In Sub-Saharan Africa with the geography, sprawl and disconnectedness in some cities (Baruah et al., 2017), a 7 km × 7 km neighborhood was the least generous smoothing window to eliminate all the 'holes'.[5] Contiguity as discussed and stated in Appendix B1 is based on the 4 rook neighbors (N, S, E and W) with a shared border, rather than the 8 queen neighbors, 4 of which just touch the own cell. The area covered by these contiguous squares over 1500 per sq. km defines what we call the core of the city. We then add in a fringe to each core, which includes all contiguous (rook) grid squares with population density over 500 per sq. km. The core combined with a fringe is called a city. For smaller urbanized places that are stand-alone, or towns, we take the collection of contiguous grid squares all with (a smoothed) population density over 500 per sq. km., which collectively sum to 5000 or more.

In comparison, the European Union Commission defines core cities based on the 1500 people per sq km and the 50,000 total population threshold. They then define towns and suburbs using 300 people per sq km and the 5000 total population cut-off. We think 300 people per sq km in some parts of the world covers very rural populations and in the discussion of Nairobi below we point out why, in defining metropolitan area suburbs, 300 is too low. For us it was more whether to cut at 500 or 750 in Sub-Saharan Africa. Also most critically for us, the EU program does not define metropolitan areas, only core cities. "Suburbs" are unassigned to core cities, which means we could not have metropolitan areas as a unit of analysis.

The process and impact of threshold decisions are illustrated in Fig. 1 for Nairobi. Core city areas are shaded in the darkest grey (almost black), and overall cities are outlined in dark blue and have corresponding stripes. There are two cities in the figure, Machakos to the bottom right and Nairobi. Nairobi consists of the main core and three small core areas, essentially satellite towns now falling under the umbrella of Nairobi. The fringe of the city of Nairobi consists of two shaded areas: an area one grey tone lighter than the core (> 750 people per sq km) and and area two tones lighter (> 500 people per sq km). Our choice of 500 per sq. km is based on the idea that a lower threshold such as 300 per sq. km (the lightest grey) is too loose and extends too far into more rural and low-density towns much further north of Nairobi. And it would place the centroid of Nairobi well outside its true central core. A higher cutoff of, say, 750 people per sq. km. (one shade darker than the core) may be too stringent and exclude satellite cities around Nairobi that are very likely to be within the commuting zone. Obviously, other arguments about drawing boundaries can be made. In the figure we also outline in light yellow color the polygons (with yellow stripes) that make up the independent towns around Nairobi. Some are

[5] See de Bellefon et al. (2018) on similar issues defining cities with built cover data.

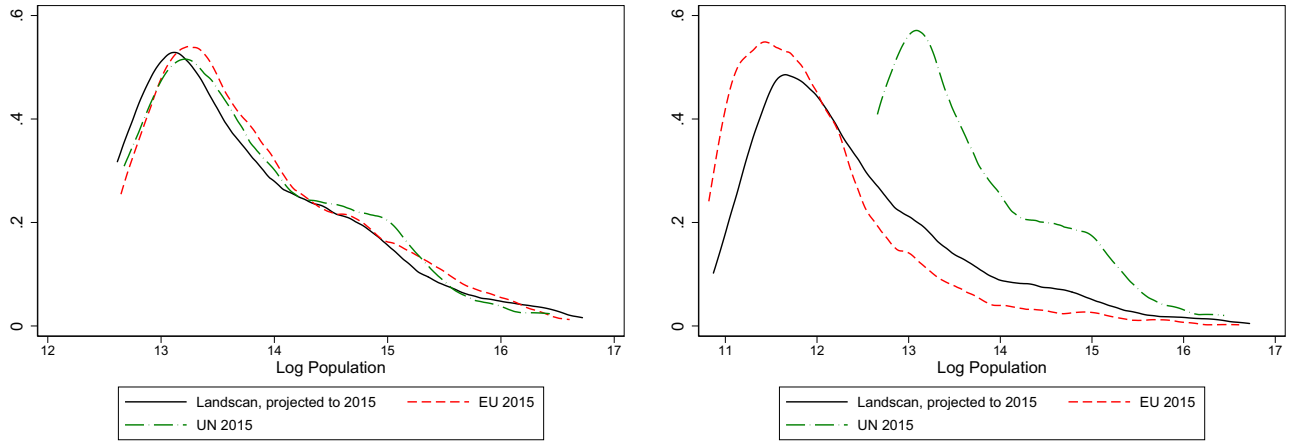


Fig. 2. Distribuitions of the city sample in the African continent according to three data sources.

very spatially distinct, but some follow ribbons (roads) to the north outside Nairobi, where rural areas are interspersed with urbanized towns. In the figure everything in lightest grey or the Google Earth background is rural. In the on-line Appendix in Figure B2 we do a version where we show the county administrative boundaries. Restricting Nairobi to be its own county would miss significant portions of population even within the core city. Adding the county to the north would add in much of the missing population but would also add in large tracks of land with low intensity of use and result in significant mismeasurement of economic land.

In Fig. 2 we compare the distribution of cities by size in our data with those of the UN and those based on the GHSL. In this comparison the city numbers (179) and extents are based on Landscan data and our algorithm. We then populate these areas with Landscan data, GHSL data and UN data, with the last based on UN cities which fall within our boundaries, noting that UN cities are cut at 300,000. Thus we are comparing how much population each source allocates to given area. In Fig. 2(a), to compare with the UN, we cut our cities at 300,000. The size distributions are pretty interwoven, although as Fig. 2(b) reveals there are differences. Fig. 2(b) keeps the UN cut, but adds in all our Landscan cities, reducing the city size cut for Landscan and GHSL numbers to 50,000, to see what happens if we populate all our algorithm cities with Landscan versus the latest version of GHSL data. The interesting feature is the shift right in the Landscan distribution, implying that for cities in general, Landscan tends to put more ambient population into the same dense urban areas as compared to GHSL.

3. Defining economic density

What is the most meaningful measure of economic density? It should be a density measure capturing the most relevant aspects of proximity among people in a city. That is, the proximity that would be most relevant for agglomeration benefits, such as cheaper trade, improved hiring and learning, and a wider choice of inputs and products. A large city population may not always generate proximity gains. A city that occupies a large area may also have very low density and low proximity, compared to a more compact city with the same population. In the same way, a city's high average density may not mean there is very high proximity of activity within the city.

Fig. 3 illustrates this last issue about proximity.[6] All hypothetical cities in Fig. 3 have the same total population (180) and average population density [PD] (5). City 1 has no clustering. Cities 2 and 3 have the same degree of within grid square clustering, with half the grid squares with no population and half with 10 people per grid. The 10 means greater within grid square possibilities for intersecting with others (the pairwise possibilities for meetings for example, n (n-1)/2, or 45 in this case). However city 2 allows for more possibilities for interactions with neighbors. Ignoring the boundaries in city 3, on average a grid square has 40 queen neighbors, while in city 2 a grid square has 80 queen neighbors in the surrounding 8 squares.

We now turn to two measures which reflect these differences relative to PD, personal population density [PPD] and a measure based on De La Roca and Puga (2017) which we label RPA in honor of the authors. For a given city area, relative to population density, personal population density is a weighted, rather than simple, sum of own cell population densities, where the weights are a cell's share of the city population. So in Fig. 3, PPD gives a value of 5 for city 1 and 10 for cities 2 and 3. The De La Roca and Puga (2017) based measure further makes a distinction between cities 2 and 3. A city-wide measure of RPA is a sum of grid square measures, where each measure is a distance discounted sum of your own and neighbors density within a given radius. Each grid square measure is weighted by its population share in the city. This measure by incorporating neighbors will give a higher value for city 2 than 3.

More generally, for personal population density [PPD] the measure for city j with N_j grid squares, i, is a weighted average of grid square populations P_{ij}, with weights being the grid squares' shares of city population, P_j. That is,

$$PPD_j = \sum_i^{N_j} P_{ij}\frac{P_{ij}}{P_j} = PD_j\left(1+\frac{Var(P_{ij})}{PD_j^2}\right) = PD_j(1+CV(P_{ij})^2) \qquad (1)$$

where CV is the coefficient of variation and $PD_j = \frac{\sum_i^{N_j} P_{ij}}{N_j}$. PPD can thus be decomposed into overall population density [PD], a typical scale measure, and one plus the coefficient of variation squared. The latter captures the degree of variation relative to the mean within the city and thus the degree to which activity is concentrated within particular cells. So cities 2 and 3 in Fig. 3 (ignoring city bounds) have the same degree of variation and clustering, and both are higher than city 1.

Note that the coefficient of variation has a long history, starting from Williamson (1965), for use as a measure of regional income inequality within a country. Here we are using it as a measure of economic density inequality within a city or town. Of course, urban economics has other measures of spatial inequality including spatial Hirschman-Herfindahl indices [HHI] and Gini's. We focus on the coefficient of variation because it comes from the natural decomposition in Eq. (1); and

[6] Fig. 3 and the definition and decomposition of personal population density are borrowed from on-going work by Henderson, Storeygard and Weil. This joint intellectual ownership is gratefully acknowledged.

City 1

5	5	5	5	5	5
5	5	5	5	5	5
5	5	5	5	5	5
5	5	5	5	5	5
5	5	5	5	5	5
5	5	5	5	5	5

City 2

0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10

City 3

0	10	0	10	0	10
10	0	10	0	10	0
0	10	0	10	0	10
10	0	10	0	10	0
0	10	0	10	0	10
10	0	10	0	10	0

Fig. 3. Differences in city layout and density measures.

this decomposition which carries over in essence to the De La Roca and Puga (2017) measure.

The RPA measure (De La Roca and Puga, 2017) is still a weighted average, but now of the own gird square plus discounted neighbor cell populations over a given radius from the own grid square. The measure and its decomposition are

$$RPA_j = \sum_i A_{ij}\frac{P_{ij}}{P_j} = AD_j\left(1 + \frac{Cov(A_{ij}, P_{ij})}{AD_j PD_j}\right) \quad (2)$$

where $AD_j = \frac{\sum_i^{N_j} A_{ij}}{N_j}$; $A_{ij} = \sum_{k\epsilon s} P_{kj}e^{-\alpha d_{ik}}$. In Eq. (2), A_{ij} is the measure over radius s of the discounted sum of neighbors ambient populations. The discount rate we use is 0.7 and is derived empirically in Section 4. The neighborhood, s, of a grid cell is defined as the square area running 5 cells to the east, west, north and south of the own cell, or an area of size 11x11 grid squares (which would be similar to a circle of radius 6.2 km). We limit the local radius so we can distinguish later the effects of city wide versus local density. AD_j is the simple average of the A_{ij} across grid squares over the city. RPA_j is the weighted average of the A_{ij}, where the weights are each grid squares share of the city population. RPA_j can then be decomposed into the simple average, and 1 plus the covariance of A_{ij} and P_{ij}, divided by their simple averages. The latter term captures the degree to which population is allocated to grid squares with high measures of neighbors (city 2 in Fig. 3), as opposed to either being uniformly spread (city 1) or being in grid squares which are not clustered with others of high density (city 3). Note, the personal population density measure in Eq. (1) is a special case of the De La Roca and Puga (2017) measure in Eq. (2), when radius s is set to 0.

We will have measures of PD_j, PPD_j and RPA_j at the level of a city or town. We will also have local measures, characterizing the neighborhood around which people live both for rural and urban areas, including for neighborhood i, PPD_{ij}, PD_{ij}, and A_{ij}. These we will describe in the particular contexts in which they arise. In all cases, the local neighborhood of a grid cell is the square area running 5 cells to the east, west, north and south of the own cell, or an area of size 11x11 km.

Most of the basics of what we have presented are not our invention. Small and Cohen (2004) calculate, on a coarser scale, a spatial Gini as a measure of within-city variation in activity. Implicit in the De La Roca and Puga (2017) measure is the notion of personal population density (when s is set to zero). Moreover, we borrowed the De La Roca and Puga (2017) measure we use, although from Section 4 we add an optimally derived spatial discount factor. They in fact set that discount factor to 0. The advantage of a non-zero discount factor is that far away places within a city get less weight, which makes the setting of city boundaries less critical for the key interior parts of a city. What we add which we believe to be novel, based on-going work by Vernon Henderson, Adam Storeygard and David Weil, are decompositions for the personal population density and De La Roca and Puga (2017) measures.

4. How are differences in economic density across the spatial hierarchy related to income differences?

This section first describes the data on income and wages and then the characteristics of the sample of cities and towns in the covered countries. After giving the base specification, we turn to a set of results on the relationship between agglomeration measures and income and wages, covering all areas of the country. In the next section, we will delve into looking at scale effects for cities in particular.

4.1. The data and the sample of countries and cities

We use the Living Standards Measurement Study data of the World Bank, where we have detailed geocoding of where families live for six countries; allowing us to map data to our spatial units: rural, towns and cities. The LSMS surveys have detailed and consistent data at the household and individual levels on income, education, labor allocation, asset ownership, and dwelling characteristics. The data sets are the Tanzania Panel Household Survey (2008 and 2010), the Nigeria National Household Survey (2010 and 2012), the Uganda National Panel Survey (2009, 2010, 2011, and 2012), the Ethiopia Socioeconomic Survey (2011, 2013, and 2015), the Malawi Integrated Household Survey (2010 and 2013), and the Ghana Socioeconomic Panel Survey (2010 and 2013). Note that the dates of surveys in countries are so close together that they do not provide the opportunity to look at dynamics nor to identify urbanization effects off of movers.[7] These sample countries account for approximately 35% of the subcontinent's population.

Before proceeding we note aspects of our Sub-Saharan African countries' urban hierarchy and the coverage of this hierarchy by LSMS surveys. We combine the urban data on the six countries and show that collectively they have a regular urban hierarchy. As a basic descriptive, Fig. 4 shows approximately a log-normal distribution of all urbanized areas including both cities and towns similar to that in Eeckhout (2004) for the USA, although there is a right tail skew. Also as a descriptive, we looked at the rank size rule for the collective, although the rank size rule should be defined as country-specific. As in Eeckhout (2004) for the

[7] There is an issue of the same households appearing more than once in our data, which varies from country to country. For a sample of 44,140 households, there are 23,685 unique households, meaning that 46% of the sample involves a household that is included more than once. Clustering at the local area should remove the distortion this creates. As a robustness check, we reran basic tables with just the final year sample in the year of the LSMS for each country. Results are very similar, with similar statistical significance and coefficient magnitudes.

Table 1
Counts of urbanised areas in our countries and sample.

	All urban areas in our countries			Urban Areas where surveyed units live			
Country	Cities	Cores	Towns	Cities	Cores	Fringes	Towns
Ethiopia	24	32	244	71%	63%	21%	14%
Ghana	15	15	66	67%	53%	40%	23%
Malawi	5	5	52	80%	80%	80%	50%
Nigeria	96	129	493	61%	49%	32%	9%
Tanzania	24	24	71	63%	50%	33%	21%
Uganda	15	15	139	100%	93%	93%	40%
Totals	179	220	1065	67%	55%	38%	18%

Notes: The table shows the counts of urban areas by type in the seven countries, and the share of them populated by the Living Standard Measurement Surveys.
Source: World Bank Living Standard Measurement Surveys and Landscan (2012).

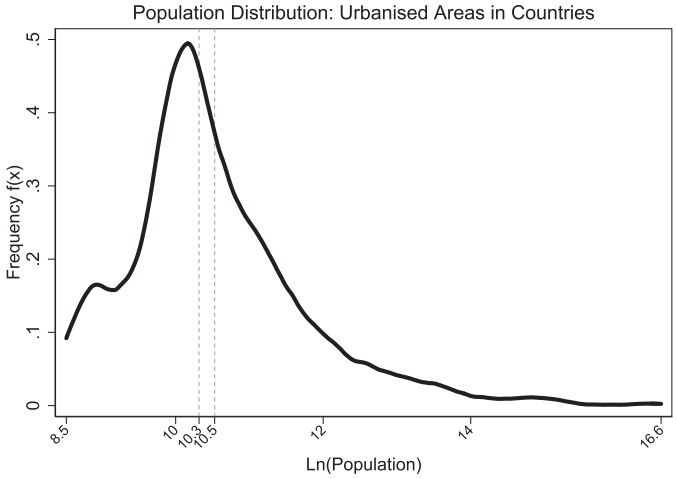


Fig. 4. Overall size distribution of urbanized areas.

USA, the rank size rule is approximated in the right tail of Fig. 4 which displays like a Pareto distribution.

How complete is the LSMS coverage of this hierarchy? Table 1 shows the distribution of cities with their cores and fringes broken out for our countries and for the LSMS sample. The left part of the table tells us that these countries have 179 cities (and fringes), covering 220 cores; and they have 1065 towns, apart from rural areas. The right part of the table shows that the LSMS data covers 67% of the 179 cities; but within these cities, only 38% of fringe areas are covered. And for towns only 18% of the 1065 are covered. The relatively low count of small places actually surveyed comes from the randomized sampling procedure outlined in Appendix B.

How representative is coverage by the LSMS? Fig. 5 breaks the urban sample in Fig. 4 into cities and towns. It compares the size distributions for all cities and for all towns in the six countries, with the size distributions of the cities and town that are covered by the LSMS. The shapes of distributions of both cities and towns for the sample are similar to those for the countries overall. The mean and median sizes for each distribution are each marked with dotted lines, with the mean being bigger than the median. For towns, the means and medians in the LSMS sample are larger than for the whole sample, and the same is the case for cities. This of course is consistent with Table 1 and the sampling procedure.

Next, we look at the characteristics of households in the sample. Table 2 gives characteristics of the LSMS households (top panel) and working age people (bottom panel) in the sample by our spatial units. Education of the household head and working-age population decline pretty sharply as we move down the spatial hierarchy. Rural areas, towns, and fringes of cities are much more likely to have the household head or workers in agriculture than the core. Virtually no one anywhere is in manufacturing, the big issue for Sub-Saharan African cities (Henderson and Kriticos, 2018). Even the proportions in business services, which are potentially tradeable across cities, are not that high, at 9% for cities and 2% in rural areas for household heads. Business services include the usual business service industries such as real estate and finance but add in high skill workers (like managers) in retail, as well as senior administrators and high skill workers in government. Apart from agriculture, even in cities, it seems that most Sub-Saharan Africans work in low skill retail services and general labor services. However, a key issue with this industry data is that many people and household heads do not report an industry. Based on IPUMS data (Henderson and Kriticos, 2018), we believe this may occur because many of these people are farmers with agricultural land who work in other sectors as well, and so don't have one industry with which they identify. We note this non-reporting fraction is noticeably higher at well over 50% in rural areas. In specifications below we will control for industry fixed effects, distinguishing from not recorded agriculture, business services, construction, education and health, labor services, manufacturing, utilities, and retail and wholesale.

Finally, there are the income measures. We focus on total household income and then separately wage rates for individuals working for wage income. We construct measures of income for the household by adding together all income from self-employment, labor income, and capital or land income. In the surveys, respondents report income receipts of various forms, such as cash and in-kind wage payments, business incomes, remittances, incomes from the rent of property and farmland, private and government pensions, and sales revenue from agricultural produce. These receipts are also reported as taking place over a variety of time intervals, so to be consistent, we convert all income receipts to monthly intervals. Land income from crop sales or rents is generally only available at the household level, making it difficult to ascribe these income sources to any particular household member for an individual-level analysis. And the same comment applies to non-agricultural businesses owned by the household head or others in the family. Therefore, for individual-level analysis we only use individual wages. More details on the calculation of net income and hourly wages are provided in Appendix B.2.

4.2. Basic specification and urban-rural results

Our first regressions have the following general specification:

$$ln(y_{ijzft}) = \alpha X_{ijzft} + \beta I_Z + \gamma_R I_R * E_{ijRf} + \gamma_S I_S * E_{ijSf} + \gamma_C I_C * E_{ijCf} + \delta \xi_{ft} + \epsilon_{ijzft} \quad (3)$$

- $ln(y_{ijzft})$: Income of household unit or hourly wage of person i in location j of type z in country f at time t.
- X_{ijzft}: Vector of household unit, person, or city characteristics.
- I_Z: Vector of indicators of location type: rural[R], town[S], city[C].
- E_{ijRf}: Measure of rural scale within a 6km radius of unit.
- E_{ijSf}: Measure of town scale.
- E_{ijCf}: Measure of city scale.
- ξ_{ft}: Vector of country-year FEs
- ϵ_{ijzft}: Error term.

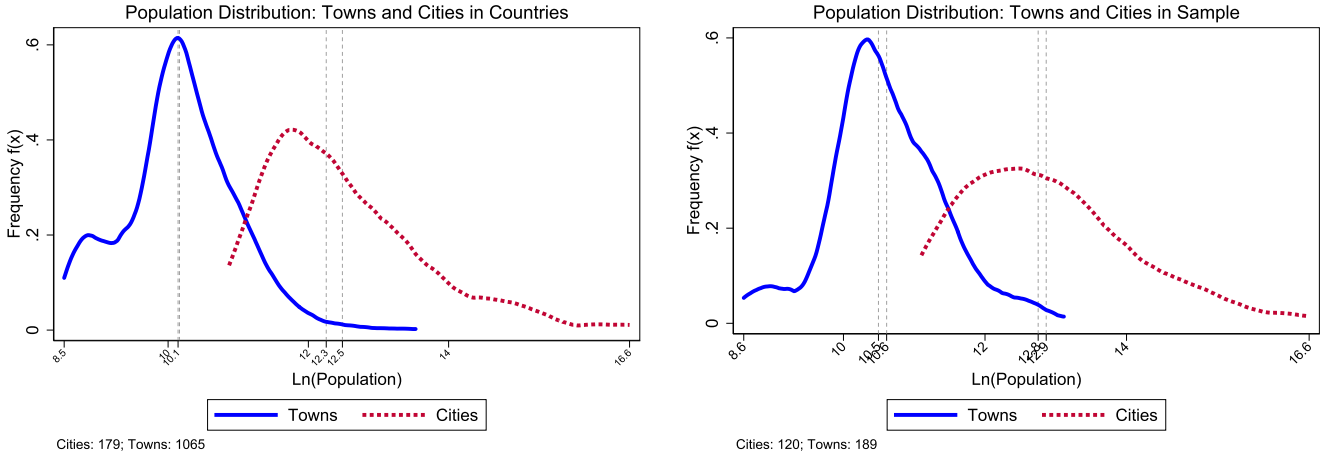


Fig. 5. Size distribution of urbanized areas.

Table 2
Household and person characteristics by location.

	City	Core	Fringe	Town	Rural
Panel A					
Household head in Agriculture	0.124	0.0629	0.273	0.261	0.297
Household head in Business Services	0.0885	0.108	0.0418	0.0510	0.0238
Household head in Manufacturing	0.0271	0.0297	0.0205	0.0220	0.00877
Household head in Not-Recorded	0.362	0.375	0.329	0.397	0.561
Household head with > Primary Education	0.441	0.503	0.290	0.307	0.173
Panel B					
Worker in Agriculture	0.0904	0.0404	0.214	0.239	0.281
Worker in BS	0.0608	0.0738	0.0286	0.0309	0.0164
Worker in Manufacturing	0.0246	0.0265	0.0199	0.0248	0.0148
Worker in Not-Recorded	0.467	0.473	0.452	0.473	0.579
Worker with > Primary Education	0.5	0.544	0.391	0.332	0.198
No. Urban Areas	120	121	68	189	–

Notes: The table shows the shares of occupations in our sample out of 1) the sample of household heads and 2) working age population (18–60 years old).
Source: World Bank Living Standard Measurement Surveys and Author's dataset.

In the first specification we will not have scale measures, just indicators to get town and city household income or wage premiums over rural areas. Then we will add in population and density scale measures. In the next section we will drop rural areas and towns and just look at cities, so the indicator terms and non-relevant scale measures will disappear.

What we estimate in this cross-section are correlations of income with scale measures, based on within country and year variation. We cannot claim causal effects for two reasons. First, there is the issue of sorting by unobserved 'ability' across space, although that has been downplayed in some of the literature (Baum-Snow and Pavan, 2011), especially once one controls for education and age. De La Roca and Puga (2017) find that OLS cross-section results perform better than some other specifications such as individual fixed effects in panel data. For sorting, an issue is whether to control for occupation or industry fixed effects as a way of trying to factor in ability conditional on education. Here we have industry fixed effects. Controlling for these fixed effects does reduce magnitudes and to be conservative we rely on these estimates, although we show some results without these controls. By doing so, however, we eliminate the scale benefit of greater choice of industries and occupations as we move up the urban hierarchy. The second issue in terms of identification is that bigger cities may have unobserved attributes which, apart from the scale, enhance productivity, such as local public infrastructure investments. But for that, at least, the estimates will give a sense of the income pull force of cities even if scale externality effects themselves could be over (or under) stated.

Table 3 presents preliminary results, which give the town and city household income and hourly wage rate premiums above the rural sector. Total household income and hourly wage rates are our two outcome measures throughout. We note that results for total wage income with the same controls are very similar to those for wage rates. Columns 1 and 2 look at effects on household income for 43,214 households, without and then with industry fixed effects for the household head. The controls include education, gender, age and age squared of the household head, household size and its square, whether the household owns land and, if so, the amount owned. Columns 3 and 4 look at hourly wages for the 19,901 individuals who work for wage income, again without and then with industry fixed effects of the individual. The controls include education, gender, age and age squared, as well as a cubic in hours worked. The coefficients on controls are reported in Appendix A for the specifications in Table 5 below.

In Table 3, row 1 gives the premium from being in a town compared to a rural area ranging from 33% to 15%. Row 2 gives the premium to being in a city compared to rural area ranging from 72% to 27%. The wage premiums at 33 and 27% for being in large city are larger but comparable to the 25% often quoted number for the USA from Glaeser and Mare (2001). What is new is the much larger premium for household incomes, at least in the context of Sub-Saharan Africa. We note of course that these are nominal income not consumption differences. To know how real incomes rise we would need cost of living or detailed consumption data, for which we have neither.

There are three clear patterns. First, being in a city compared to a town gives a much bigger premium, over twice as much for household incomes and just under twice as much for wages rates. Second, adding in industry fixed effects reduces premiums by up to 30%. While in the

Table 3
Gains from urban type.

	(1) HH income	(2) HH income	(3) Hrly wage	(4) Hrly wage
Town	0.333***	0.213***	0.194***	0.145***
	(0.0314)	(0.0301)	(0.0309)	(0.0307)
City (core + fringe)	0.720***	0.524***	0.326***	0.270***
	(0.0224)	(0.0225)	(0.0190)	(0.0193)
Observations	43,214	43,214	19,901	19,901
R^2	0.335	0.383	0.296	0.313
Country-Year FE	√	√	√	√
Controls	√	√	√	√
Occupation FE		√		√

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Notes: Table reports results from OLS regressions; the dependent variable in columns (1)-(2) is the natural logarithm of total net monthly income from all available sources with the mean of 3.88. The dependent variable in columns (3)-(4) is the natural logarithm of hourly wage income with the mean of 1.21. Controls at the household level are the education (recorded or not), level of education (if recorded), age, age squared and gender of the household head; household size; household size squared; whether the household owns land and, if so, the natural logarithm of the size of land holdings in hectares. Controls at the individual level are education (recorded or not); level of education (if recorded); age; age squared; gender; and hours worked squared and cubed. The sample in columns (3)-(4) is limited to individuals aged 18 to 60, working for a wage and reporting hours worked. Each column includes country-year fixed effects. Columns (2) and (4) include industry fixed effects for household head and individual worker respectively. Robust standard errors are presented in parentheses. Asterisks indicate $p < 0.01$(***), $p < 0.05$(**), and $p < 0.1$(*). *Source: Author's dataset*

rest of the paper we include industry fixed effects as a control on unobserved attributes of people, we still worry that biases the returns to scale downwards by ignoring industry-occupation diversity and upgrading effects. Finally, returns to household income from moving up the spatial hierarchy are higher than for wages, a pattern we will focus on below. In column 2 the 52% premium to being in a city over a rural area for household income is about twice that for wages at 27% in column 4.

In the second set of preliminary results in Table 4, we turn to specifications where we experiment in each column with a different measure of economic density, proceeding from total population, to population density [PD], personal population density [PPD] and finally the De La Roca and Puga (2017) based measure [RPA] defined in equation (2). In panel A, we look at household incomes, with controls for household and household head characteristics including industry fixed effects. For each spatial group– rural, town, and city– we estimate a marginal scale effect for the four measures of scale. For rural, the scale measure applies to the 11x11 km square around the households grid square. In all columns in Table 4 we allow the income intercept to vary by spatial type, so as to normalize scale starting points relative to rural areas (like 50,000 for population for cities). But our interest is in marginal scale effects.

For rural households, local scale has no positive impact on household incomes. For towns population effects are negative but density effects are very high: an elasticity of 0.43 for mean density down to 0.11 for personal population density. For cities, the unit people usually focus on, the population elasticity is 0.085, but the density ones range from 0.38 to 0.54. A population elasticity of 0.085% is considerably higher than the often quoted numbers like 0.02-0.03 (e.g., from Combes et al., 2008) for developed countries. In Panel A, going from a rural area to a city of 5 million population raises household incomes by 55% (0.55 log points from 0.0846*15.4 - 0.75). However, in comparison, the returns to density are extraordinary. Within just cities, going from the lowest to the highest PD, PPD, or RPA raises household incomes by 1.14–1.52 log points (see Table A1 on minimum and maximum values of each measure for the city sample). We will do detailed comparisons to the literature on other developing countries in the next section where we focus just on density measures for the sample of cities and explain the divergence between population and density returns to scale. We do note here that for developed countries, in the classic paper by Ciccone and Hall (1996), doubling density raises productivity by 6%; here doubling density raises household incomes by 37% and in Panel B raises wage rates by 10%, as discussed below.[8]

In terms of which density measure is either sufficient or better, we explore that in the next section looking just at cities. We note here that, for just cities, the density elasticities are comparable across measures. While elasticities decline as we move from PD (0.52) to PPD (0.38) to RPA (0.37), their standard deviations within the city sample rise from 0.61 to 0.71, to 0.84 respectively. Between towns and cities, elasticities for PD are similar; but for PPD and to some extent for RPA, they are higher for cities.

So far we have mostly discussed household income effects in Panel A. Panel B examines hourly wage effects. The first comment is that corresponding to Table 3, scale elasticities for wages tend to be much smaller than for household incomes. For cities for density measures, marginal scale effects for household incomes tend to be about 3–4 times higher than for wages; and the population elasticity for wages for cities is zero. That said, wage density effects for cities are strong. Going from the lowest to highest PD, PPD, or RPA within the city sample raises hourly wages by over 30%.

5. Economic density in cities

In this section, we will delve into looking at scale effects in cities in particular and focus on the extraordinary density effects we have uncovered in Sub-Saharan Africa. The first question is why simple population and density measures yield so different elasticities. The second concerns which economic density measure suffices or does better than the others in explaining income and wage differences: population, density, personal population density, the RPA measure and components of the latter two. Before proceeding we need to tell how we derive the discount factor used in the RPA measure, used in part in assessing the degree to which within city clustering matters.

5.1. Constructing de la Roca-Puga measures

We take the specification in column 4 of Panel A of Table 4 and drop all observations outside of cities and thus the rural and town scale terms and the dummies for town and city, so we are left with just the RPA measure for cities plus our basic controls. To recall, the RPA in effect measures how many people a person taken at random in a city can reach. A more realistic version of that measure would discount the number of people within reach according to how far away they are. What discount rate should we use, recognizing that such a discount rate is context specific and depends on local transport infrastructure and culture. We approach this problem empirically to get an overall rate for our sample. We start with a spatial discount rate of −0.1 and raise that in absolute value in increments of 0.1 to −1.5. One should note that −1.5 is an extremely high discount: At 1 km distance, neighbors have a weight of 0.22; at 2 km it is already only 0.05, and by 5km their weight is effectively 0. For each discount rate, we record the F-value of adding the ln RPA term; these values are shown in Figure B3 in the on-line Appendix. The peak is in the neighborhood of −0.7, so that the improvement in explaining income differences across cities is maximized around a −0.7 discount rate. We do note values from −0.5 to −0.9 yield similar Fs under different specifications. We use the discount rate of −0.7 in all cases for any type of RPA measure. This discount rate indicates that neighbors,

[8] The recent evidence from De La Roca and Puga (2017) finds even smaller earnings elasticity for city density in Spain of 0.045; and the authors do not find sizable effects of sorting on the unobservables that would affect the measured city premium.

Table 4
Gains from scale by area type.

Panel A. Household incomes				
	(1) Log Population	(2) Log Mean Density	(3) Log PPD	(4) Log RPA
Rural X Scale	−0.00979	−0.0141	0.00379	−0.00712
	(0.0105)	(0.0108)	(0.00790)	(0.0101)
Town X Scale	−0.0857***	0.429***	0.107**	0.196**
	(0.0313)	(0.147)	(0.0526)	(0.0763)
City (core + fringe) X Scale	0.0846***	0.540***	0.440***	0.377***
	(0.0111)	(0.0261)	(0.0223)	(0.0189)
Town	1.044***	−2.657***	−0.610	−1.643**
	(0.352)	(0.972)	(0.429)	(0.716)
City (core + fringe)	−0.752***	−3.541***	−3.508***	−3.584***
	(0.184)	(0.202)	(0.211)	(0.218)
Observations	43,214	43,214	43,192	43,214
R^2	0.384	0.389	0.389	0.389
Panel B. Individual hourly wage premiums				
	(1) Log Population	(2) Log Mean Density	(3) Log PPD	(4) Log RPA
Rural X Scale	−0.0324**	−0.0298**	0.00732	−0.0315**
	(0.0141)	(0.0146)	(0.0105)	(0.0133)
Town X Scale	−0.0226	0.0875	0.186***	0.164**
	(0.0315)	(0.143)	(0.0557)	(0.0812)
City (core + fringe) X Scale	0.00684	0.143***	0.106***	0.110***
	(0.00838)	(0.0207)	(0.0187)	(0.0155)
Town	0.0882	−0.572	−1.334***	−1.619**
	(0.367)	(0.946)	(0.462)	(0.768)
City (core + fringe)	−0.127	−0.945***	−0.672***	−1.163***
	(0.176)	(0.173)	(0.186)	(0.196)
Observations	19,901	19,901	19,899	19,901
R^2	0.314	0.315	0.315	0.315

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Notes: Table reports results from OLS regressions. The dependent variable in Panel A is the natural logarithm of total net monthly income from all available sources with the mean of 3.88. The dependent variable in Panel B is the natural logarithm of hourly individual wage income with the mean of 1.21. Scale variable varies by column and is indicated in column headings. Controls are the same as in Table 3. Each column includes country-year and industry fixed effects. The sample in Panel B is limited to individuals aged 18 to 60, working for a wage and reporting hours worked. Robust standard errors are presented in parentheses. Asterisks indicate $p < 0.01$(***), $p < 0.05$(**), and $p < 0.1$(*). *Source: Author's dataset*

in general, are not so important. With this discount, at 2km, neighbors only get a weight of 0.25 and at 5km a weight of 0.03.

5.2. Economic density results for cities

We start by asking why population and density elasticities are so different, what measures best capture economic density, and how our wage results compare to the literature. Then we will discuss why household income premiums in denser cities are much larger than individual wage premiums.

Our basic results are in Table 5, where we look just at people living in cities. In columns 1, 3 4 and 5 we first repeat what in essence are the columns 1–4 results in Table 4 on returns to overall density measures. Any any differences in specific results between the tables arise from having just a city sample for which the effects of controls are estimated, as well as adding two new variables. These are meant to capture issues of how market access can affect income and wages, where market access may be also correlated with density and thus constitutes an omitted variable (Hanson, 2005). We control for the distance from each city centre to the nearest port (Geospatial Geoscience Ltd, 2017) and we add a measure of market access defined as

$$MA_i = \sum_{j=1}^{N} POP_j * e^{-0.002d} \quad (4)$$

POP_j is population of a neighboring city j within 3000 km. We apply a distance discount $e^{-0.002d}$, where d is distance. There is no robust trade elasticity for the African continent in the literature. We chose the parameter 0.002 in the weight as it matches a reasonable discount. For example, a city 100 km away loses around 18% of its population and one 1000 km away loses 86% of its population, which seem plausible trade losses in Sub-Saharan Africa.

Adding the market access controls and changing the sample in estimating the effect of controls make little difference for the city results. For example, the population and density elasticites in Table 5 in columns 1 and 3 are 0.0506 and 0.523 for household incomes and 0 and 0.169 for wage rates; these are similar to those in Table 4 of 0.0846, 0.54, 0, and 0.143 respectively. Table 5 is the one which in Appendix A we show two other things: coefficients on all controls Tables A5 and A6 and the quite similar scale effects if we omit industry fixed effects (Table A2). Tables A5 and A6 also show the very significant effects of market access and distance to ports on wages and incomes; but these controls have essentially no impact of density returns.[9]

[9] We also note that results are very similar for all other countries if we drop Nigeria, the largest country in the sample. The coefficients in Panel A on PD, PPD and RPA without Nigeria are 0.635, 0.386 and 0.416 (vs. respectively from Table 5 Panel A: 0.523, 0.382 and 0.372). For hourly wages they are 0.167, 0.112 and 0.120 (vs. respectively from Table 5 Panel B 0.169, 0.148 and 0.134). In general they are slightly higher for household incomes and slightly lower for wage rates if Nigeria is omitted. We also show that hourly wages results on density effects are similar for males versus females. For PD, PPD and RPA for wages only, male (female) coefficients are respectively 0.192 (0.132), 0.149 (0.148)

Table 5
Gains from different measures of scale for the city sample.

Panel A. Household incomes							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ln(Total Population)	0.0506	0.516***					
	(0.0473)	(0.0813)					
Ln(Pop. Density)			0.523***		0.531***		
			(0.0822)		(0.0830)		
Ln(PPD)				0.382***			
				(0.0838)			
Ln(1 + CV Term)					0.0918		
					(0.129)		
Ln(RPA)						0.372***	
						(0.0604)	
Ln(AD)							0.462***
							(0.0864)
Ln(1 + Cov Term)							0.159
							(0.202)
Ln area		−0.542***					
		(0.0840)					
Observations	11,237	11,237	11,237	11,237	11,237	11,237	11,237
R^2	0.314	0.330	0.330	0.326	0.330	0.330	0.330
Panel B. Individual hourly wage premiums							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ln(Total Population)	−0.00681	0.160***					
	(0.0244)	(0.0380)					
Ln(Pop. Density)			0.169***		0.180***		
			(0.0393)		(0.0387)		
Ln(PPD)				0.148***			
				(0.0369)			
Ln(1 + CV Term)					0.0609		
					(0.0625)		
Ln(RPA)						0.134***	
						(0.0291)	
Ln(AD)							0.132***
							(0.0472)
Ln(1 + Cov Term)							0.139
							(0.119)
Ln area		−0.198***					
		(0.0410)					
Observations	9358	9358	9358	9358	9358	9358	9358
R^2	0.364	0.369	0.368	0.368	0.368	0.369	0.369

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Notes: Table reports results from OLS regressions. The dependent variable in Panel A is the natural logarithm of total net monthly income from all available sources with the mean of 4.86. The dependent variable in Panel B is the natural logarithm of individual hourly wage income with the mean of 1.38. Controls are the same as in Table 3; additional controls include natural logarithm of distance to the nearest port and natural logarithm of market access to all cities and towns, weighted by $e^{(-0.002*d)}$ where d is distance. The sample in Panel B is limited to individuals aged 18 to 60, working for a wage and reporting hours worked. Each column includes country-year and industry fixed effects. Standard errors are clustered at the city level. Asterisks indicate $p < .01(***)$, $p < .05(**)$, and $p < .1(*)$. *Source: Author's dataset*

5.2.1. Simple density versus population measures

We now address the question of why population and density effects differ so much, whether for household incomes or for wage rates. The answer is that column 1 in Table 5 with ln population as the economic density measure has an omitted variable, land, which is correlated with population. When we add ln land, in column 2 the coefficient on population increases tenfold to 0.52, so conditional on land, increases in population have a huge impact. Most crucially, in column 2 when we add ln land area to column 1, for each of household incomes and wage rates, the coefficients on population and land are close to each other in absolute value and of opposite signs (0.516 versus -0.542 and 0.16 versus -0.20 for household incomes and wage rate respectively). That justifies the column 3 specification with just ln population density as the economic density measure, where we impose equal and opposite sign coefficients on population and land. This neat result of land and population having essentially the same coefficients of opposite sign occurs because we are able to measure "economic land" so precisely. What matters is population density in column 3, not population; and the density elasticity is enormous: 0.52 for household incomes and 0.17 for wages in Sub-Saharan Africa.[10]

Here a comparison with the recent literature is helpful. All the papers cited focus on wage rate outcomes and all use administrative boundaries to define land areas. Quintero and Roberts (2018) pool 16 Latin American countries much like we pool African countries. They have various

and 0.149 (0.110). For all measures point estimates are somewhat higher for males.

[10] In noting the divergence between marginal returns to population versus and density, we note that, in Sub-Saharan Africa, density and city population are not well correlated, with a simple correlation coefficient of only 0.31 (in logs). There are some very big, but low density cities. But even smaller cities can have higher density than more medium sizes ones. For personal population density, its mean for the lowest quartile of cities by size is 13,324 compared to the means of 9847 and 10,857 for the second and third quartiles by city size.

Table 6
Why household income gains are larger than for wages?

	(1) Working for wage	(2) Working for wage	(3) Hours worked for those recorded	(4) Hours worked for those recorded	(5) N of unique occupations in a household	(6) N of unique occupations in a household
Ln(Pop. Density)	0.000285	−0.00434	3.399***	3.788***	0.136***	−0.172
	(0.00470)	(0.00304)	(0.565)	(0.546)	(0.0380)	(0.201)
HH size					0.159***	−0.327
					(0.0358)	(0.278)
Ln(Pop. Density) × HH size						0.0646
						(0.0405)
Observations	10,211	4264	9358	3920	13,508	13,508
R^2	0.071	0.044	0.067	0.084	0.620	0.628
Country-Year FE	√	√	√	√	√	√
Controls	√	√	√	√	√	√
Women Only		√		√		
Mean of Y			48.22	46.69	1.258	1.258

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Notes: Table reports results from OLS regressions; the dependent variable is indicated in the column titles. Columns (1)-(4) use the sample of individuals aged 18 to 60, restricting to the sample of just women in columns (2) and (4). Column (5) presents a sample of households. Controls are the same as in Table 3. Each column includes country-year fixed effects. Standard errors are clustered at the individual urbanised area level. Asterisks indicate $p < 0.01$(***), $p < 0.05$(**), and $p < 0.1$(*).
Source: Author's dataset

Table 7
Estimation ofaverage and local density effects on household income premiums.

	(1)	(2)	(3)	(4)	(5)
Ln(City PD)	0.421***	0.462***	0.442***	0.436***	0.438***
	(0.0948)	(0.0900)	(0.0961)	(0.0846)	(0.0896)
Ln(Local PD)	0.142***		0.171***		
	(0.0438)		(0.0465)		
Ln(Local PPD)		0.162***			0.00908
		(0.0438)			(0.0684)
Ln(1 + Local CV Term)			0.137*		
			(0.0738)		
Fringe				−2.452***	−4.225***
				(0.894)	(0.999)
Fringe × Ln(City PD)				0.294**	0.272**
				(0.120)	(0.121)
Fringe × Ln(Local PPD)					0.237***
					(0.0830)
Observations	11,237	11,237	11,237	11,237	11,237
R^2	0.286	0.287	0.287	0.289	0.291
Country-Year FE	√	√	√	√	√
Controls	√	√	√	√	√
Occupation FE	√	√	√	√	√

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Notes: Table reports results from OLS regressions; the dependent variable is the natural logarithm of total net household income from all available sources with the mean of 4.86. Controls are the same as in Table 3 at the household level. Additional controls include natural logarithm of distance to the nearest port and natural logarithm of market access to all cities and towns, weighted by $e^{(-0.002*d)}$ where d is distance. Each column includes country-year and occupation fixed effects. Standard errors are clustered at the individual urbanised area level. Asterisks indicate $p < 0.01$(***), $p < 0.05$(**), and $p < 0.1$(*). *Source: Author's dataset*

estimates but at most the density elasticity is about 0.06 which corresponds to the estimate in Duranton (2016) for Colombia. Our estimate of 0.17 is much higher; but their uses of administrative boundaries create error in measurement of economic land. In Combes et al. (2019) the authors investigate several outcomes in China. For high and low skill urban natives in cities, they find higher density elasticities than Latin America ranging from 0.06 to 0.16 depending on the specification. Interestingly and conveniently, they control for ln land area as well. Given that, as one specific but representative example (Table 2, panel b, column 2) the implied population elasticity is 0.141 and the land one is -0.056, very different from our situation where with precise land measures the absolute magnitudes are almost equal. In Chauvin et al. (2017) (Table 8, first panel), effects of ln population and ln density are investigated separately for the USA MSA's, Brazil microregions, Chinese cities and India districts, all based on land measured for administrative boundaries. Density elasticities for China are 0.192 while population ones are insignificant. Density elasticities in Brazil and India are lower at 0.026 and 0.076 respectively. However, the samples include rural and town areas as well, so results not directly comparable to ours, based on our Table 4 results. For USA MSA's, the density elasticity is 0.046, much less than for Sub-Saharan African and Chinese cities. In sum, for Sub-Saharan African cities, we find large wage elasticities compared to much of the literature and get unmuddied density results given our economic land areas are precisely measured.

5.2.2. Second moment measures

Does the degree of clustering within cities matter in this developing country context? The results on within city degree of clustering are in

Table 8
Estimation of average and local density effects on individual wage premiums.

	(1)	(2)	(3)	(4)	(5)
Ln(City PD)	0.153***	0.156***	0.172***	0.112***	0.122***
	(0.0479)	(0.0415)	(0.0497)	(0.0359)	(0.0369)
Ln(Local PD)	0.0435		0.0673		
	(0.0410)		(0.0419)		
Ln(Local PPD)		0.0727**			−0.00902
		(0.0362)			(0.0408)
Ln(1 + Local CV Term)			0.0932**		
			(0.0441)		
Fringe				−2.276***	−3.349***
				(0.582)	(0.637)
Fringe × Ln(City PD)				0.291***	0.278***
				(0.0782)	(0.0719)
Fringe × Ln(Local PPD)					0.139*
					(0.0719)
Observations	9358	9358	9358	9358	9358
R^2	0.350	0.351	0.351	0.354	0.355
Country-Year FE	√	√	√	√	√
Controls	√	√	√	√	√
Occupation FE	√	√	√	√	√

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Notes: Table reports results from OLS regressions; the dependent variable is the natural logarithm ofhourly wage income with the mean of 1.38 . Controls are the same as in Table 3 at the individual level. Additional controls include log distance to the nearest port and natural logarithm of market access to all cities and towns, weighted by $e^{(-0.002*d)}$ where d is distance. The sample is limited to individuals aged 18 to 60, working for a wage and reporting hours worked. Each column includes country-year and occupation fixed effects. Standard errors are clustered at the individual urbanised area level. Asterisks indicate $p < 0.01$(***), $p < 0.05$(**), and $p < 0.1$(*). *Source: Author's dataset*

Table 5 columns 4 and 6. In column 4, we decompose ln PPD into the ln PD and ln (1 + coefficient of variation term) in equation (1). The coefficient of variation term is small and insignificant and thus does not add to explanatory power relative to just using ln PD in column 2. In column 6, we repeat the same exercise for the RPA measure getting the same result; the covariance term from equation (2) is insignificant. We also conducted a series of horse-races in Appendix Tables A3 and A4. There, a horse-race between PD and PPD suggests PD dominates for both household income and wage rates. For PD versus RPA there is no strict pattern of domination by one or the other. In short, using PD seems as good or better than using PPD or RPA, and the two measures of the differential degree of clustering within cities do not seem to add to the analysis.

There are two issues with drawing the conclusion that we should focus on PD rather than PPD or RPA measures for cities. First concerns measurement error. Use of Landscan data measures within-city clustering and inequality with error. While Landscan may do a better job than other currently available data sets, the assignment of people to work and residential locations is surely done with considerable error, which would bias the coefficients downward. The second issue concerns our measure of clustering. While our measure of clustering fits into a decomposition and is a standard measure in looking at spatial inequality, there are other standard measures. We looked at one of these. In a context with so many grid cells in each urban area but a varying number, we preferred the spatial Gini to use in comparison over an HHI or a Theil index. We calculated the spatial Gini of cell concentration of economic activity within each city.[11] The Gini had also a completely insignificant coefficient as reported in an earlier version of this paper (see Henderson et al., 2018a).

The conclusion for this sample and data is that a simple population density measure works as well as more nuanced measures, in attempting to capture *overall* economic density of cities in explaining income differences. However, later when we consider local within city density measures, our more nuanced measures will have more purchase.

5.2.3. Why are economic density effects for wage rates and household incomes so different?

What is stunning in the tables are the extraordinarily high density elasticities for household incomes compared to wages. Why might this be? We explore three channels here: greater labor force participation within the household in bigger or denser cities with better job opportunities, opportunities and choices to work longer hours, and an opportunity to upgrade and diversify occupations or industries. We look at these in Table 6. In columns 1 and 2, we look at labor force participation in terms of working for wage income or not, for those 18–60 for all and for females. Density has no effect. However, we find effects for the other two channels.

In columns 3 and 4 of Table 6, we look at hours worked for those working for wages. We find similar effects for all as we do for just females alone. A one standard deviation (0.61) increase in density raises hours worked by 2.31 from a mean of 47 for females and similarly overall (in a formulation where the count of hours is in logs, the elasticities are both 0.09). Next in columns 5 and 6 for the household we look at the count of different primary industries of employment listed by different members of the household, controlling for household sizes. So if all work in retail, the count is one. If two people work in retail and one in manufacturing the count is 2. In column 5, a one standard deviation increase in density raises the count of number of different industries of household members by 0.08 from a mean of 1.3 (and in a log formulation the elasticity is 0.063). In column 6 we see if density interacts with household size but the coefficient is insignificant.

Two channels do seem to be working: denser cities offer the opportunity and choice to work more hours and for the household member to

[11] We approximate the Gini by ordering each cell by its density and noting the cumulative share of the population in each cell. The cumulative distribution of the population ordered by density represents the Lorenz curve. We then sum up the area under the Lorenz curve (by adding up the "height" and the "width" of each bar of the cumulative population histogram, where the "height" is the cumulative population and the "width" is 1/(number of total cells in the city)). We call this integral I, and, according to the Gini formula, calculate Gini = 1-2I for each city.

match to a greater variety of industries. We note that in Table 2 there are a lot more manufacturing and business service jobs in cities than in towns or rural areas. We documented (not reported) that household activity in agriculture declines sharply with density. We think families move to cities, not just for wage gains but for huge household income gains based on the opportunities denser places offer household members to work longer hours and to better match to industries outside of agriculture for which different household members have more interest and better skills. There must be other channels, given our large differences in income and wage returns to density, which warrant investigation in the future.

5.2.4. Does neighborhood density matter?

A key issue as noted in the introduction is that studies suggest that within cities there are local scale externalities which decay sharply with distance so that firms in one neighborhood do not seem to interact with firms in another (Arzaghi and Henderson, 2008), at least in the dimensions being captured. That begs the question of why there can be multiple neighborhoods of seemingly non-interacting firms found in one city. The answer must be that firms also benefit more generally from overall urban scale. Here we show results where both overall city density and local density matter, so both elements are captured.

In Tables 7 and 8, columns 1–3 of each, we address this issue. All these columns control for citywide ln PD and then introduce a local measure for the 11 × 11 km neighborhood around the own grid square (i.e., a radius of just over 5 km). Local and city densities will tend to be positively correlated, so if the local density measures matter that will reduce the elasticity of ln city PD which is 0.52 in column 2 of Table 5(A) and 0.169 in column 2 of Table 5(B). This is definitely the case for Table 7 on household incomes but less clear for Table 8 for wages. In either case the coefficients on ln city PD are still very large, indicating huge marginal returns to overall city density. In each of columns 1–3 in Tables 7 and 8, we experiment with a different measure of local economic density. Although we have argued that a simple PD measure works as well or better than anything else for a citywide measure, that does not mean at the local level it is the best measure. In column 1 the local measure is ln (local PD); in 2 it is ln (local PPD); and in 3 we decompose local PPD as in Eq. (1) into PD and the covariance term.

For household incomes, in all three columns in Table 7, local density measures have strong significant effects; and column 3 in Table 7 suggests local covariance matters, so that local clustering within your neighborhood helps incomes. Local magnitudes are all smaller, about 0.16, than city wide PD elasticities of 0.46 in this table. Horse-races in Appendix Table A3 for household income suggest no clear winner between local PD and local PPD, although local PPD dominates local RPA (and hence we only report it in the working paper (Henderson et al., 2018a)).

In Table 8 on wages, there is less impact of local measures. Local PD is smaller and insignificant. Local PPD is significant at the 5% level, as is the local coefficient of variation measure. Quantitatively, as in Table 7, in Table 8 elasticities for local PD measures are about 1/4 to 1/3 of those for the overall city PD measure; but standard errors are too large for these smaller magnitudes to be significant. It may be that local economic density is more important for household incomes which may depend on local jobs for household members who are part time or supplemental earners and who may work more for non-wage income, than at an hourly rate.

Finally we note that these results pertain to wage rates and household incomes. Firm productivity effects may be a different story. In Appendix E we look at neighborhood effects for firms in Kampala. Having greater Landscan density in the neighborhood around a firm does nothing to productivity, while for households having greater local density raises incomes in Table 7. Firms only seem to respond to greater local (but not immediate) own industry employment density. Perhaps this differentiation for results for households versus firms is not surprising. For households it is about job search, job training, and learning (De La Roca and Puga, 2017).

In summary, for a household, the overall density measure for the city has strong effects on incomes, but also the local area around the household has strong effects as well. Households benefit from both dense overall cities and dense local neighborhoods.

5.2.5. The core versus fringe of cities

Finally, we examine how effects may differ for people living in the core versus fringe of cities, which will suggest more work for the future. We look at this issue in the last two columns of Tables 7 and 8. In both Tables 7 and 8, while those living in the fringe start from a much lower base income or wage rate (the fringe indicator), in column 4, they get an extra kick from city PD, with an elasticity of density that overall is 0.73 for household incomes (versus 0.44 for those living in the core) and 0.40 for wages (vs a much smaller 0.11 for those living in the core). These are huge differences which warrant further exploration. We note that for fringe residents it is overall city, not core PD which matters as reported in an earlier version of the paper (Henderson et al., 2018a). We looked similarly in column 5 in Tables 7 and 8 at the effects of local PD for those in the core versus the fringe. In both tables, the results suggest local PD matters just for those living in the fringe. One thought is that in the core, densities may be more uniform and higher, while at the fringe, local density may be much more variable and that is highly relevant.

The puzzle is the generally bigger role of city density externalities for fringe than core city residents. We thought this was because of sorting based on migration status, where migrants lack information and might benefit more from say information spillovers. Hence, we expected that the fringe would have a greater proportion of migrants (defined by having moved within the last 5 years) than the core within the LSMS sample. However, there is only a tiny difference: 16% vs 15%. Moreover, coefficients on migration status interacted with scale effects are insignificant and have no impact on the fringe results. So that leaves a puzzle. There certainly must be sorting on some dimension, where people in the fringe earn less for the same observables, but somehow this disadvantaged population benefits more from greater city density, even though by construction they live in less dense neighborhoods.

6. Conclusions

This paper evaluates the use of different measures of economic density in assessing urban agglomeration effects to try to carefully measure and characterize the pull force of cities. These density measures are based on Landscan data, which we argue in Appendix C does a good job in capturing within city variation in economic density. The Landscan data allow us to precisely and consistently define cities and measure the economic extent of the city, or its land area.

To assess economic density measures, we examine how well they explain household income and wage differences across cities and neighborhoods. We have simple scale and density measures and more nuanced ones which capture in second moments the extent of clustering within cities. Noting that the extent of clustering is measured with error resulting in attenuation bias, the evidence suggests that a simple, but consistently calculated density measure explains income differences across cities as well as or even better than more nuanced measures which attempt to represent within city differences in the extent and nature of clustering. However, simple city scale measures such as total population are distinctly inferior to density measures.

Overall, there are huge household income premiums from being in bigger and particularly denser cities over rural areas and small towns in Sub-Saharan Africa, indicating migration pull forces are very strong. It may be that part of urban-rural premiums are explained by low productivity in the farming sector. Additionally, the selection of unobservably abler migrants into cities may also add to the size of the effect, although we have many controls on household and member attributes. However, within the sample of cities, the marginal effects of increases in density on household income are enormous by any standard, with a density

elasticity of 0.52. Besides overall city density measures, we look at density in the neighborhood around a household within a city: in addition to strong city-level density effects we find strong neighborhood effects looking at neighborhoods of about 6km radius. The elasticity of overall city density is 0.42 and for local density it is 0.14. Both overall city density and density of the own neighborhood matter.

For wages, while density premiums are on the high end relative to the literature, they are much lower than for household incomes. We found two channels for why income elasticities are higher, although more must be at work: people work longer hours as density rises and households diversify industries.

Future research would reestablish current results with the new datasets on density soon to emerge, and some of which are at a much more disaggregated scale (Worldpop (100 m), High Resolution Human Settlement Layer (30 m), MAUPP (12.5 m)). Higher resolution data on all aspects will uncover more accurate clustering variation within cities, to help us establish how important that is for households' earning potential. Adding detailed within-city transport information that is consistent across cities can also paint a better picture on the true proximity between workers; see Akbar et al. (2018) for steps in this direction. Having economic census data would allow better assessment of different forms of density– diurnal versus nocturnal. Individual panel data might help in identification and uncovering of dynamic effects, but such panels are in their infancy anywhere in Africa.

Published Appendix A: Statistics and other results

Table A1
Summary statistics.

Panel A: Cities

Variable	Mean	Min	Median	Max	STD	N
Local measures						
Ln(Local Landscan Linear Sum (5km))	12.13	10.03	12.02	14.45	1.05	11,237
Ln(Local PD)	7.78	5.77	7.65	10.17	1.07	11,237
Ln(Local PPD)	9.17	6.34	9.29	10.92	0.97	11,237
Ln(Local RPA)	10.39	8.22	10.28	12.79	1.10	11,237
Citywide measures						
Ln(Total Population)	14.06	11.10	14.44	16.58	1.43	11,237
Ln(Pop. Density)	7.47	6.51	7.34	8.65	0.61	11,237
Ln(PPD)	9.30	7.36	9.30	10.81	0.71	11,237
Ln(Coefficient of Variation)	1.83	0.81	1.81	3.04	0.37	11,237
Ln(RPA)	10.87	9.26	10.64	12.28	0.84	11,237
Ln(RPA) unweighted	9.88	8.95	9.73	11.13	0.60	11,237
Ln(Cov. Term RPA)	1.00	0.26	0.99	1.88	0.31	11,237

Panel B: Towns

Variable	Mean	Min	Median	Max	STD	N
Ln(Total Population)	11	9	11	13	1	3440
Ln(Pop. Density)	6.57	5.86	6.55	7.19	0.18	3440
Ln(PPD)	8.10	6.27	8.12	9.33	0.52	3440
Ln(RPA)	9.31	8.63	9.28	10.11	0.34	3440

Panel C: Rural

Variable	Mean	Min	Median	Max	STD	N
Ln(Local Landscan Linear Sum (5km))	9.06	0.00	9.18	11.52	0.97	28,537
Ln(Local PD)	4.68	0.00	4.80	7.12	0.93	28,537
Ln(Local PPD)	6.05	0.37	6.16	9.89	1.26	28,515
Ln(Local RPA)	7.18	0.00	7.30	9.42	0.99	28,537

Note: The table presents summary statistics of the natural logarithms of the city-level and local variables of interest. Panel A is the summary for the sample of just cities, Panel B is the summary of town-level measures for towns, and finally, the Panel C is the summary of the local measures for rural survey locations.

Table A2
Gains from scale by area type. No industry fixed effects.

Panel A: Household incomes				
	(1) Log Population	(2) Log Mean Density	(3) Log PPD	(4) Log RPA
Rural X Scale	−0.00735	−0.0129	0.0163*	−0.00247
	(0.0109)	(0.0113)	(0.00830)	(0.0105)
Town X Scale	−0.0860***	0.546***	0.192***	0.299***
	(0.0329)	(0.155)	(0.0549)	(0.0798)
City (core + fringe) X Scale	0.0723***	0.558***	0.456***	0.392***
	(0.0114)	(0.0264)	(0.0228)	(0.0192)
Town	1.189***	−3.304***	−1.111**	−2.458***
	(0.370)	(1.024)	(0.447)	(0.749)
City (core + fringe)	−0.363*	−3.481***	−3.388***	−3.535***
	(0.190)	(0.207)	(0.218)	(0.224)
Observations	43214	43214	43192	43214
R^2	0.336	0.341	0.341	0.341
Panel B: Individual hourly wage premiums				
	(1)	(2)	(3)	(4)
Rural X rursc	−0.0204	−0.0182	0.0175*	−0.0171
	(0.0142)	(0.0146)	(0.0106)	(0.0133)
Town X Scale	−0.0265	0.0350	0.171***	0.142*
	(0.0319)	(0.144)	(0.0568)	(0.0827)
City (core + fringe) X Scale	0.00397	0.157***	0.113***	0.119***
	(0.00844)	(0.0208)	(0.0187)	(0.0156)
Town	0.292	−0.122	−1.093**	−1.261
	(0.372)	(0.956)	(0.471)	(0.782)
City (core + fringe)	0.0815	−0.939***	−0.621***	−1.096***
	(0.177)	(0.174)	(0.187)	(0.197)
Observations	19,901	19,901	19,899	19,901
R^2	0.296	0.298	0.297	0.298

Notes: Table reports results from OLS regressions; the dependent variable in Panel A is the natural logarithm of total monthly net income from all available sources with the mean of 3.88. The dependent variable in Panel B is the natural logarithm of hourly individual wage income with the mean of 1.21. Scale variable varies by column and is indicated in column headings. Controls are the same as in Table 3. Each column includes country-year. The sample in Panel B is limited to individuals aged 18 to 60, working for a wage and reporting hours worked. Robust standard errors are presented in parentheses. Asterisks indicate $p < 0.01$(***), $p < 0.05$(**), and $p < 0.1$(*). *Source: Author's dataset*

Table A3
Horserace: Estimation of density effects on household incomes.

	(1)	(2)	(3)	(4)
Ln(Pop. Density)	0.439***	0.303	0.407***	0.418***
	(0.132)	(0.224)	(0.0893)	(0.0859)
Ln(PPD)	0.0918			
	(0.129)			
Ln(RPA)		0.167		
		(0.170)		
Ln(Local PD)			0.0483	
			(0.0657)	
Ln(Local PPD)			0.0964	0.105*
			(0.0680)	(0.0545)
Ln(Local RPA)				0.0360
				(0.0497)
Observations	11,237	11,237	11,237	11,237
R^2	0.330	0.330	0.332	0.332
Country-Year FE	√	√	√	√
Controls	√	√	√	√
Occupation FE	√	√	√	√

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Notes: Table reports results from OLS regressions; the dependent variable is the natural logarithm of total net monthly income from all available sources with the mean of 4.86. Controls are the same as in Table 3 at the household level. Additional controls include log distance to the nearest port and natural logarithm of market access to all cities and towns, weighted by $e^{-0.002*d}$ where d is distance. Each column includes country-year and industry fixed effects. Standard errors are clustered at the city level. Asterisks indicate $p < 0.01$(***), $p < 0.05$(**), and $p < 0.1$(*). *Source: Author's dataset*

Table A4
Horserace: Estimation of density effects on individual hourly wage premiums.

	(1)	(2)	(3)	(4)
Ln(Pop. Density)	0.119*	0.0117	0.156***	0.149***
	(0.0687)	(0.138)	(0.0441)	(0.0369)
Ln(PPD)	0.0609			
	(0.0625)			
Ln(RPA)		0.126		
		(0.105)		
Ln(Local PD)			−0.0315	
			(0.0523)	
Ln(Local PPD)			0.0802**	0.0754**
			(0.0402)	(0.0365)
Ln(Local RPA)				−0.0253
				(0.0413)
Observations	9358	9358	9358	9358
R^2	0.368	0.369	0.369	0.369
Country-Year FE	√	√	√	√
Controls	√	√	√	√
Occupation FE	√	√	√	√

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Notes: Table reports results from OLS regressions; the dependent variable is the natural logarithm of individual hourly wage income with the mean of 1.38. Controls are the same as in Table 3 at the individual level. Additional controls include log distance to the nearest port and natural logarithm of market access to all cities and towns, weighted by $e^{-0.002*d}$ where d is distance. The sample is limited to individuals aged 18 to 60, working for a wage and reporting hours worked. Each column includes country-year and industry fixed effects. Standard errors are clustered at the city level. Asterisks indicate $p < 0.01$(***), $p < 0.05$(**), and $p < 0.1$(*). *Source: Author's dataset*

Table A5
Estimation of scale effects on household income premiums.

	(1)	(2)	(3)	(4)	(5)	(6)
Ln(Total Population)	0.0412					
	(0.0530)					
Ln(Pop. Density)		0.577***		0.586***		
		(0.0887)		(0.0896)		
Ln(PPD)			0.428***			
			(0.0900)			
Ln(Coefficient of Variation)				0.101		
				(0.140)		
Ln(RPA)					0.407***	
					(0.0644)	
Ln(RPA) unweighted						0.508***
						(0.0913)
Ln(Cov. Term RPA)						0.170
						(0.223)
Education is recorded	−0.385***	−0.346***	−0.360***	−0.346***	−0.348***	−0.344***
	(0.124)	(0.122)	(0.123)	(0.122)	(0.123)	(0.123)
Kindergarten education	0.0295	0.0657	0.0442	0.0635	0.0787	0.0704
	(0.148)	(0.139)	(0.141)	(0.139)	(0.139)	(0.139)
Primary education	0.360***	0.351***	0.336***	0.346***	0.336***	0.343***
	(0.124)	(0.116)	(0.118)	(0.115)	(0.114)	(0.114)
Secondary education	0.822***	0.753***	0.768***	0.751***	0.752***	0.753***
	(0.127)	(0.118)	(0.118)	(0.117)	(0.116)	(0.116)
Specialised education	1.624***	1.530***	1.551***	1.528***	1.527***	1.530***
	(0.168)	(0.162)	(0.162)	(0.162)	(0.162)	(0.161)
Age HH head	0.00241	0.00175	0.00380	0.00214	0.00280	0.00217
	(0.00733)	(0.00717)	(0.00689)	(0.00700)	(0.00706)	(0.00709)
(Age HH head) Squared	−0.0000615	−0.0000452	−0.0000662	−0.0000486	−0.0000546	−0.0000486
	(0.0000817)	(0.0000789)	(0.0000769)	(0.0000776)	(0.0000783)	(0.0000787)
Gender	0.188***	0.197***	0.213***	0.202***	0.202***	0.200***
	(0.0612)	(0.0552)	(0.0551)	(0.0539)	(0.0554)	(0.0547)
HH size	0.146***	0.153***	0.153***	0.153***	0.151***	0.151***
	(0.0172)	(0.0165)	(0.0173)	(0.0166)	(0.0171)	(0.0168)
(HH size) Squared	−0.00269***	−0.00299***	−0.00299***	−0.00301***	−0.00290***	−0.00290***
	(0.000941)	(0.000879)	(0.000947)	(0.000888)	(0.000922)	(0.000903)
HH has land	−0.824***	−0.635***	−0.663***	−0.627***	−0.632***	−0.636***
	(0.155)	(0.119)	(0.110)	(0.114)	(0.115)	(0.117)
Ln(Total size of landholdings (ha))	0.208*	0.217**	0.209**	0.216**	0.223**	0.215**
	(0.116)	(0.102)	(0.0996)	(0.100)	(0.0982)	(0.0997)
Ln(Distance to port)	−0.0308*	−0.0285***	−0.0299**	−0.0281***	−0.0303***	−0.0299***
	(0.0177)	(0.00945)	(0.0118)	(0.00954)	(0.0104)	(0.00973)
Ln(Market Access settlement (discount = 0.002))	0.642**	0.791***	0.759***	0.798***	0.901***	0.842***
	(0.278)	(0.250)	(0.289)	(0.255)	(0.280)	(0.263)
Constant	−7.332	−13.73***	−12.86**	−14.12***	−15.74***	−15.49***
	(4.947)	(4.425)	(5.188)	(4.567)	(4.923)	(4.663)
Observations	11,237	11,237	11,237	11,237	11,237	11,237
R^2	0.263	0.283	0.278	0.284	0.283	0.283

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6
Estimation of scale effects on hourly wage premiums.

	(1)	(2)	(3)	(4)	(5)	(6)
Ln(Total Population)	−0.0152					
	(0.0254)					
Ln(Pop. Density)		0.201***		0.210***		
		(0.0448)		(0.0451)		
Ln(PPD)			0.169***			
			(0.0429)			
Ln(Coefficient of Variation)				0.0502		
				(0.0636)		
Ln(RPA)					0.155***	
					(0.0337)	
Ln(RPA) unweighted						0.164***
						(0.0506)
Ln(Cov. Term RPA)						0.131
						(0.122)
Education is recorded	−0.162	−0.152	−0.168	−0.156	−0.163	−0.161
	(0.172)	(0.173)	(0.174)	(0.174)	(0.174)	(0.174)
Kindergarten education	−0.129	−0.129	−0.128	−0.129	−0.126	−0.127
	(0.124)	(0.124)	(0.125)	(0.124)	(0.125)	(0.124)
Primary education	0.184*	0.184*	0.181	0.183*	0.183*	0.183*
	(0.110)	(0.109)	(0.109)	(0.109)	(0.109)	(0.109)
Secondary education	0.578***	0.556***	0.567***	0.557***	0.560***	0.559***
	(0.127)	(0.125)	(0.126)	(0.125)	(0.126)	(0.125)
Specialised education	1.238***	1.206***	1.223***	1.208***	1.209***	1.209***
	(0.176)	(0.174)	(0.175)	(0.174)	(0.175)	(0.175)
Age HH head	0.0817***	0.0821***	0.0826***	0.0823***	0.0822***	0.0822***
	(0.0123)	(0.0120)	(0.0120)	(0.0120)	(0.0119)	(0.0120)
(Age HH head) Squared	−0.000790***	−0.000789***	−0.000796***	−0.000791***	−0.000789***	−0.000789***
	(0.000161)	(0.000157)	(0.000157)	(0.000158)	(0.000156)	(0.000156)
Gender	0.518***	0.522***	0.522***	0.523***	0.520***	0.521***
	(0.0522)	(0.0525)	(0.0526)	(0.0525)	(0.0525)	(0.0522)
Total hrs Worked per week × Total hrs Worked per week	−0.000615***	−0.000632***	−0.000630***	−0.000633***	−0.000633***	−0.000633***
	(0.0000604)	(0.0000601)	(0.0000602)	(0.0000601)	(0.0000604)	(0.0000604)
Total hrs Worked per week × Total hrs Worked per week × Total hrs Worked per week	0.00000461***	0.00000473***	0.00000472***	0.00000474***	0.00000474***	0.00000474***
	(0.000000671)	(0.000000672)	(0.000000670)	(0.000000672)	(0.000000673)	(0.000000674)
Ln(Distance to port)	−0.0126	−0.00801	−0.00693	−0.00753	−0.00777	−0.00783
	(0.0105)	(0.00772)	(0.00838)	(0.00754)	(0.00752)	(0.00743)
Ln(Market Access settlement (discount = 0.002))	0.390*	0.457**	0.474**	0.466**	0.517***	0.508**
	(0.198)	(0.179)	(0.191)	(0.183)	(0.183)	(0.199)
Constant	−6.704*	−9.610***	−9.991***	−9.936***	−10.83***	−10.74***
	(3.529)	(3.093)	(3.415)	(3.249)	(3.243)	(3.373)
Observations	9358	9358	9358	9358	9358	9358
R^2	0.344	0.350	0.349	0.350	0.350	0.350

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jue.2019.103188.

References

Akbar, P.A., Couture, V., Duranton, G., Ghani, E., Storeygard, A., 2018. Mobility and congestion in urban India. The World Bank.
Angel, S., Blei, A.M., Jason, P., Lamson-Hall, P., Sánchez, N.G., Civco, D.L., Lei, R.Q., Thom, K., 2016. Atlas of Urban Expansion 2016 Edition, Vol. 1. New York University, Lincoln Land Institute, and UN-Habitat.
Arzaghi, M., Henderson, J.V., 2008. Networking off Madison avenue. Rev. Econ. Stud. 75 (4), 1011–1038.
Barrios, S., Bertinelli, L., Strobl, E., 2006. Climatic change and rural–urban migration: the case of sub-saharan africa. J. Urban Econ. 60 (3), 357–371.
Baruah, N.G., Henderson, J.V., Peng, C., 2017. Colonial legacies: shaping african cities. London School of Economics and Political Science, SERC DP 0226.
Baum-Snow, N., Pavan, R., 2011. Understanding the city size wage gap. Rev. Econ. Stud. 79 (1), 88–127.
de Bellefon, M.-P., Combes, P.-P., Duranton, G., Gobillon, L., Gorin, C., 2018. Delineating urban areas using building density. Processed, University of Pennsylvania.
Brückner, M., 2012. Economic growth, size of the agricultural sector, and urbanization in africa. J. Urban Econ. 71 (1), 26–36.
Bustos, P., Caprettini, B., Ponticelli, J., 2016. Agricultural productivity and structural transformation: evidence from brazil. Am. Econ. Rev. 106 (6). 1320–65
Chauvin, J.P., Glaeser, E., Ma, Y., Tobio, K., 2017. What is different about urbanization in rich and poor countries? cities in {Brazil, {China, {India and the {United {States. J. Urban Econ.cs 98, 17–49.
Ciccone, A., Hall, R.E., 1996. Productivity and the density of economic activity. Am. Econ. Rev. 86 (1), 54–70.
Collier, P., Jones, P., 2016. Transforming Dar es Salaam into a City that Works. Oxford University Press.
Collier, P., Jones, P., Spijkerman, D., 2018. Cities as engines of growth: evidence from a new global sample of cities. Unpublished.
Combes, P.-P., Démurger, S., Li, S., Wang, J., 2019. Unequal migration and urbanisation gains in china. J. Dev. Econ. forthcoming.
Combes, P.-P., Duranton, G., Gobillon, L., 2008. Spatial wage disparities: sorting matters!. J. Urban Econ. 63, 723–742.
Combes, P.-P., Gobillon, L., 2015. The Empirics of Agglomeration Economies. In: G. Duranton, J., Strange, W. (Eds.), Handbook of regional and urban economics, 5. Elsevier, pp. 247–348.
Desmet, K., Gomes, J., Ortuño-Ortín, I., 2018. The geography of linguistic diversity and the provision of public goods. National Bureau of Economic Research, Working Paper 24694.
Dingel, J.I., Miscio, A., Davis, D.R., 2019. Cities, lights, and skills in developing economies. J. Urban Econ.. In press
Duranton, G., 2015. Growing Through Cities in Developing Countries, 30. The World Bank Research Observer, pp. 39–73.
Duranton, G., 2016. Agglomeration effects in Colombia. J. Reg. Sci. 56 (2), 210–238.
Eeckhout, J., 2004. Gibrat's law for (all) cities. Am. Econ. Rev. 94 (5), 1429–1451.
Esch, T., Bachofer, F., Heldens, W., Hirner, A., Marconcini, M., Palacios-Lopez, D., Roth, A., Üreyen, S., Zeidler, J., Dech, S., et al., 2018. Where we live a summary of the achievements and planned evolution of the global urban footprint. Remote Sens. 10 (6), 895.
Fay, M., Opal, C., 1999. Urbanization without growth: a not-so-uncommon phenomenon. The World Bank, Policy Research Working Paper.
Fujita, M., Ogawa, H., 1982. Multiple equilibria and structural transition of non-monocentric urban configurations. Reg. Sci. Urban Econ. 12 (2), 161–196.
Geospatial Geoscience Ltd, 2017. World port index. https://www.arcgis.com/home/item.html?id=dd8823d9502e48c89058fc8f2c4e96ba.
Glaeser, E.L., Mare, D.C., 2001. Cities and skills. J. Labor Econ. 19 (2), 316–342.
Gollin, D., Lagakos, D., Waugh, M.E., 2013. The agricultural productivity gap. Q. J. Econ. 129 (2), 939–993.
Gollin, D., Parente, S.L., Rogerson, R., 2007. The food problem and the evolution of international income levels. J. Monet. Econ. 54 (4), 1230–1255.
Hanson, G.H., 2005. Market potential, increasing returns and geographic concentration. J. Int. Econ. 67 (1), 1–24.
Harari, M., 2016. Cities in bad shape: Urban geometry in India. Processed, Wharton School of the University of Pennsylvania.
Heblich, S., Redding, S.J., Sturm, D.M., 2018. The making of the modern metropolis: evidence from London. National Bureau of Economic Research, Working Paper 25047.
Henderson, J.V., Kriticos, S., 2018. The development of the African system of cities. Ann. Rev. Econ. 10, 287–314.
Henderson, J.V., Nigmatulina, D., Kriticos, S., 2018a. Measuring urban economic density. CEP Working Paper CEPDP1569.
Henderson, J.V., Storeygard, A., Deichmann, U., 2017. Has climate change driven urbanization in Africa? J. Dev. Econ. 124, 60–82.
Lall, S.V., Henderson, J.V., Venables, A.J., 2017. Africa's Cities: Opening Doors to the World. World Bank Publications.
LandScan, 2012. Oak Ridge National Laboratory. https://landscan.ornl.gov/.
Matsuyama, K., 1992. Agricultural productivity, comparative advantage, and economic growth. J. Econ. Theory 58 (2), 317–334.
OECD, 2012. Redefining Urban: A New Way to Measure Metropolitan Areas. Organisation for Economic Cooperation and Development (OECD).
Quintero, L.E., Roberts, M., 2018. Explaining spatial variations in productivity: Evidence from Latin America and the Caribbean. The World Bank, Policy Research Working Paper WPS8560.
De La Roca, J., Puga, D., 2017. Learning by working in big cities. Rev. Econ. Stud. 84 (1), 106–142.
Rose, A.N., Bright, E.A., 2014. The LandScan Global Population Distribution Project: current state of the art and prospective innovation. Technical Report. Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States).
Rosenthal, S.S., Strange, W.C., 2004. Evidence on the nature and sources of agglomeration economies. In: Henderson, J., Thisse, J. (Eds.), Handbook of Regional and Urban Economics, 4. Elsevier, pp. 2119–2171.
Rosenthal, S.S., Strange, W.C., 2008. The attenuation of human capital spillovers. J. Urban Econ. 64 (2), 373–389.
Small, C., Cohen, J., 2004. Continental physiography, climate, and the global distribution of human population. Current Anthropol. 45 (2), 269–277.
UN, 2015. World Urbanization Prospects: The 2014 Revision. United Nations Department of Economics and Social Affairs, Population Division: New York, NY, USA.
Venables, A.J., 2018. Urbanisation in developing economies: building cities that work. REGION 5 (1), 91–100.
Williamson, J.G., 1965. Regional inequality and the process of national development: a description of the patterns. Econ. Dev. Cultural Change 13 (4, Part 2), 1–84.